

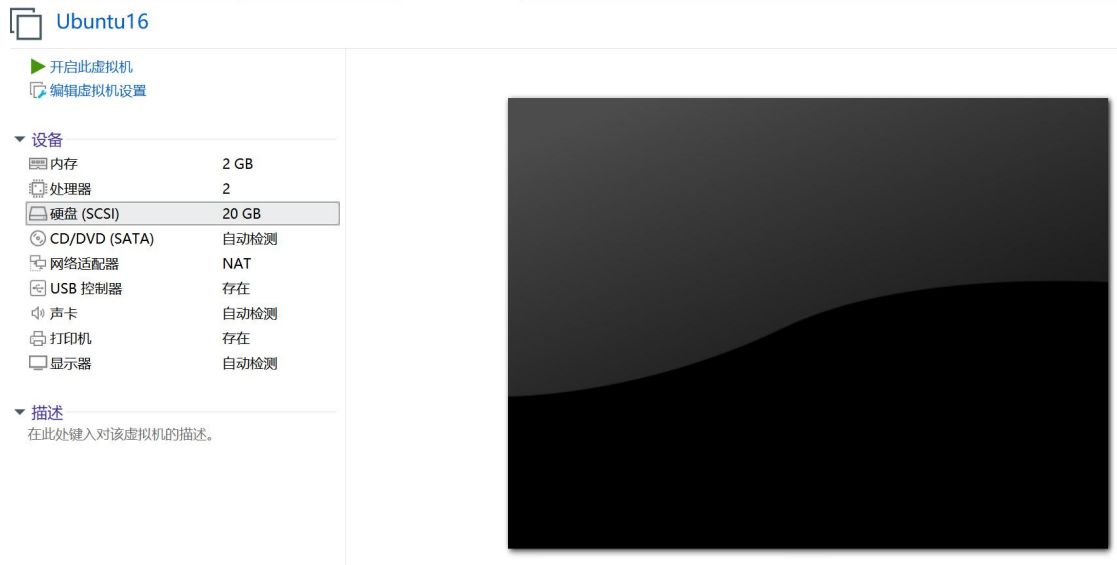
虚拟机添加硬盘分区格式化

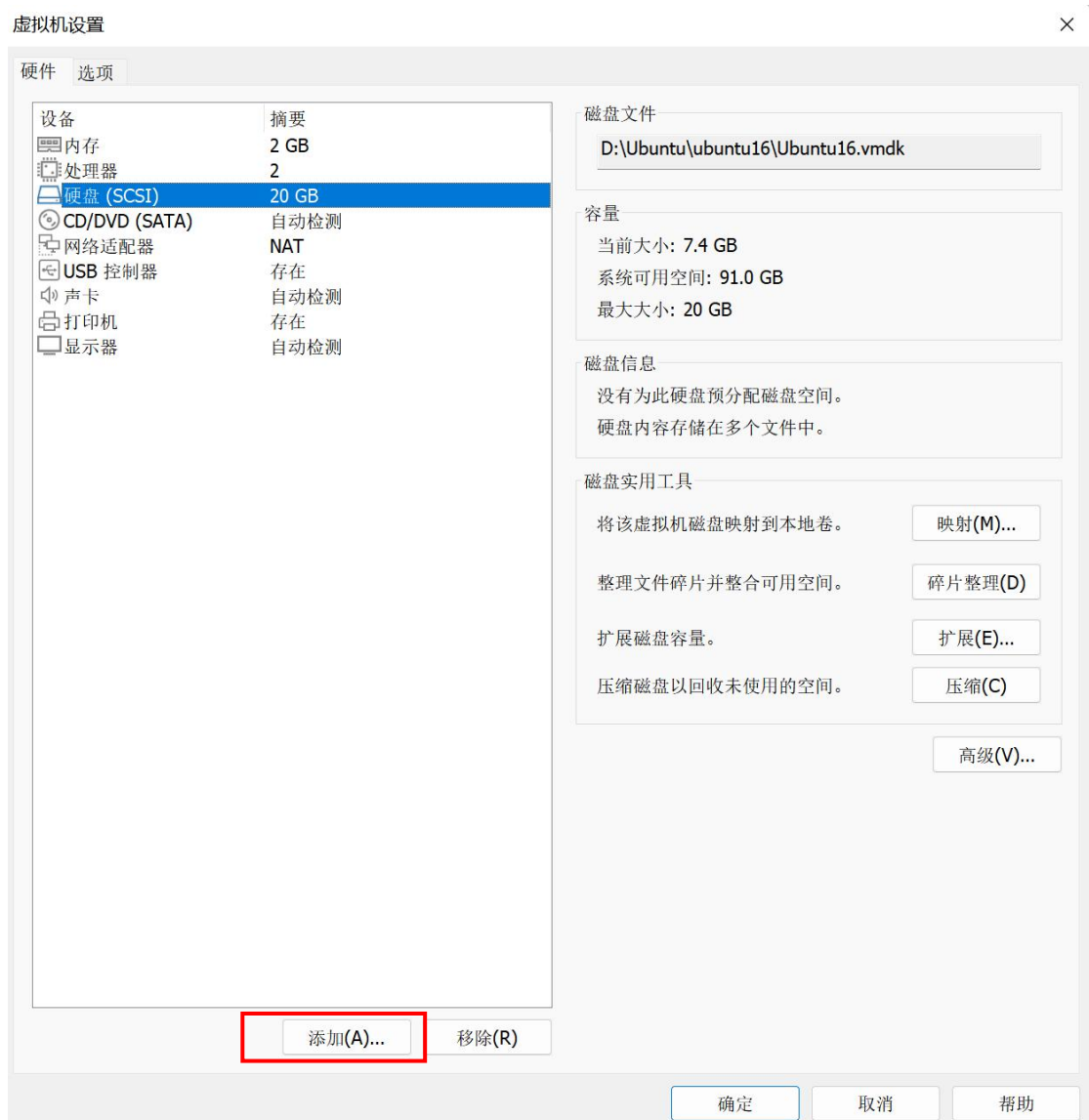
一、添加磁盘

1. Linux 系统关机

2. 虚拟机添加硬盘

找到需要添加硬盘的虚拟机系统（注意断电！添加硬盘之前不要启动机器），
点击编辑虚拟机设置





点击添加，默认都是下一步，注意指定磁盘容量时默认是 20G，这里改为 5G，点击确定。



添加硬件向导

选择磁盘

您要使用哪个磁盘？

磁盘

☒ 创建新虚拟磁盘(V)

虚拟磁盘由主机文件系统上的一个或多个文件组成，客户机操作系统会将其视为单个硬盘。虚拟磁盘可在一台主机上或多台主机之间轻松复制或移动。

☐ 使用现有虚拟磁盘(E)

选择此选项可重新使用以前配置的磁盘。

☐ 使用物理磁盘 (适用于高级用户)(P)

选择此选项可为虚拟机提供直接访问本地硬盘的权限。需要具有管理员特权。

< 上一步(B)

下一步(N) >

取消

添加硬件向导

指定磁盘容量

磁盘大小为多少？

最大磁盘大小 (GB)(S): 5

针对 Ubuntu 的建议大小: 20 GB

☐ 立即分配所有磁盘空间(A)

分配所有容量可以提高性能，但要求所有物理磁盘空间立即可用。如果不立即分配所有空间，虚拟磁盘的空间最初很小，会随着您向其中添加数据而不断变大。

☐ 将虚拟磁盘存储为单个文件(O)

☒ 将虚拟磁盘拆分成多个文件(M)

拆分磁盘后，可以更轻松地在计算机之间移动虚拟机，但可能会降低大容量磁盘的性能。

< 上一步(B)

下一步(N) >

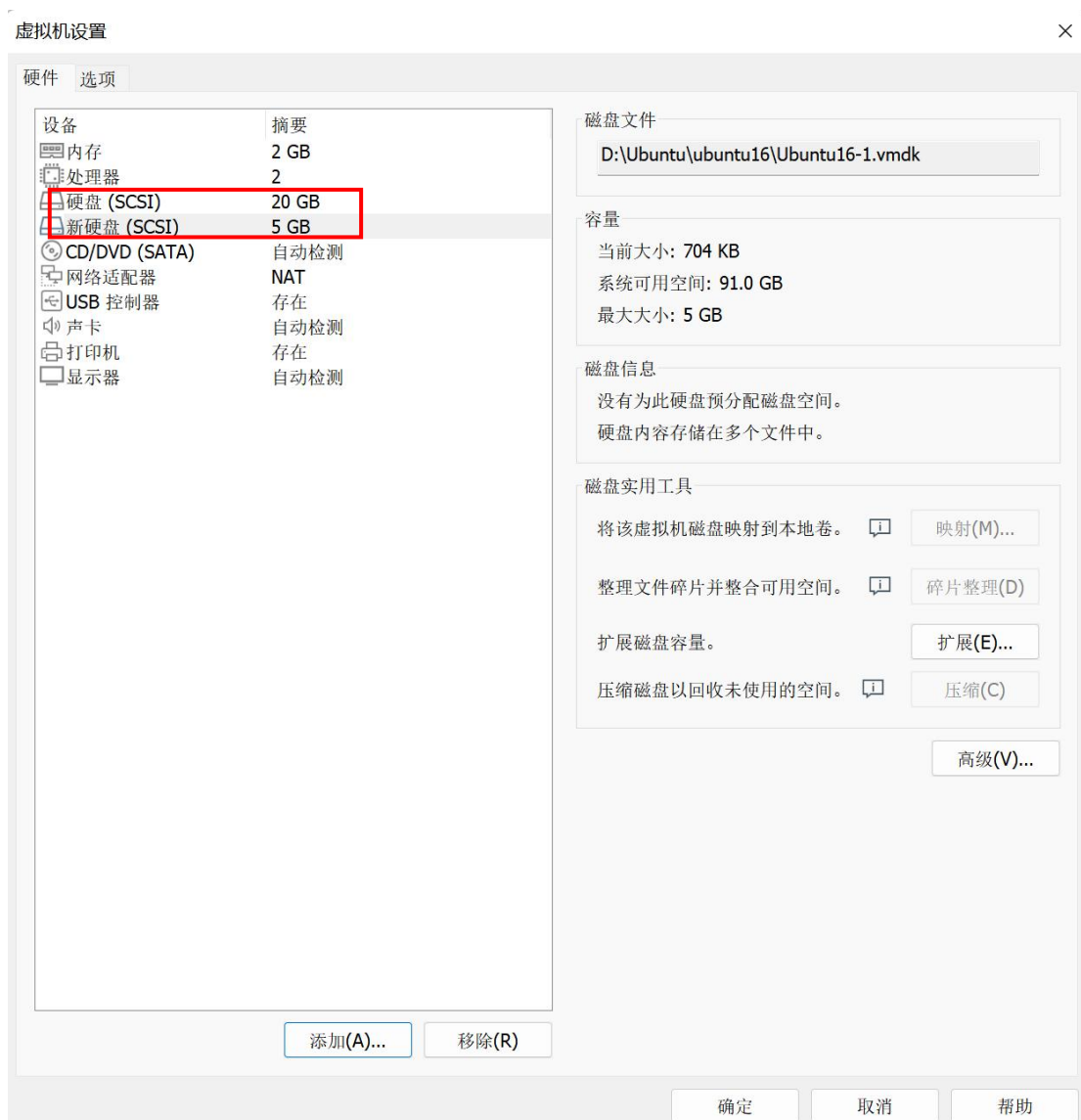
取消

指定虚拟硬盘的位置



3. 查看

打开虚拟机，此时可以看到硬盘已经添加成功，但是现在还不能使用，需要进一步配置。



启动虚拟机，使用 `ll /dev/sd*` 查看所有磁盘信息，红框中的即是新添加的硬盘。也可以使用 `lsblk` 来查看。

```
mylinux@ubuntu:~$ ll /dev/sd*
/dev/sda /dev/sda1 /dev/sda2 /dev/sda5 /dev/sdb
```

```
mylinux@ubuntu:~$ lsblk
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sdb        8:16   0    5G  0 disk
sr0       11:0    1 1024M  0 rom
sda        8:0    0   20G  0 disk
├─sda2      8:2    0    1K  0 part
├─sda5      8:5    0   975M  0 part [SWAP]
└─sda1      8:1    0   19G  0 part /
```

二、磁盘分区

使用 fdisk -l 查看磁盘信息

```
mylinux@ubuntu:~$ sudo fdisk -l
Disk /dev/sda: 20 GiB, 21474836480 bytes, 41943040 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xb0eabb67

Device      Boot    Start        End    Sectors    Size Id Type
/dev/sda1   *        2048    39942143    39940096    19G 83 Linux
/dev/sda2                39944190    41940991    1996802    975M  5 Extended
/dev/sda5                39944192    41940991    1996800    975M 82 Linux swap / Solaris

Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

使用 MBR 创建分区，输入 fdisk /dev/sdb 命令启动分区工具，进入会话模式，输入 n 开始创建分区。

- ①输入 n 后会提示要划分主分区还是扩展分区，这里选择 p 主分区；
- ②接下来选择分区号，默认从 1 号开始划分；
- ③之后是选择磁盘开始的扇区 2048-10485759，默认为 2048，这里按回车键即可；
- ④接下来是选择磁盘分区结束的扇区，即分区大小，直接输入分区的大小即可；
- ⑤输入 w 保存分区信息；

刷新分区表，使用 partprobe /dev/sdb 命令。

分区规划：

主分区 1：500M

主分区 2：1G

主分区 3：1G

扩展分区 4：剩余

逻辑分区 5：1G

逻辑分区 6：1.5G（剩余）

1. fdisk 命令对磁盘/dev/sdb 进行分区


```
mylinux@ubuntu:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.27.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xaa11e55c.

Command (m for help): m
```

输入 m，查看帮助

Generic

```
d   delete a partition
F   list free unpartitioned space
l   list known partition types
n   add a new partition
p   print the partition table
t   change a partition type
v   verify the partition table
i   print information about a partition
```

Save & Exit

```
w   write table to disk and exit
q   quit without saving changes
```

Create a new label

```
g   create a new empty GPT partition table
G   create a new empty SGI (IRIX) partition table
o   create a new empty DOS partition table
s   create a new empty Sun partition table
```

输入 n，新建一个分区

```
Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-10485759, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-10485759, default 10485759): +500M

Created a new partition 1 of type 'Linux' and of size 500 MiB.
```

分区类型：p（主分区）

分区编号：默认值，回车

开始扇区号：默认值，回车

最后扇区号：输入+500M，需要输入大写 M

输入 p 查看分区表

```
Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device            Boot Start      End Sectors  Size Id Type
/dev/sdb1          2048 1026047 1024000   500M 83 Linux
```

继续输入 n，建立第二个分区，同第一个分区建立过程，前边选项回车，分区结束扇区输入+1G，输入 p 查看分区表

```
Select (default p):
Using default response p.
Partition number (2-4, default 2):
First sector (1026048-10485759, default 1026048):
Last sector, +sectors or +size{K,M,G,T,P} (1026048-10485759, default 10485759): +1G

Created a new partition 2 of type 'Linux' and of size 1 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device            Boot Start      End Sectors  Size Id Type
/dev/sdb1          2048 1026047 1024000   500M 83 Linux
/dev/sdb2          1026048 3123199 2097152    1G 83 Linux
```

继续输入 n，建立第三个分区，同第一个分区建立过程，前边选项回车，分区结束扇区输入+1G，输入 p 查看分区表

```

Command (m for help): n
Partition type
  p   primary (2 primary, 0 extended, 2 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3):
First sector (3123200-10485759, default 3123200):
Last sector, +sectors or +size{K,M,G,T,P} (3123200-10485759, default 10485759): +1G

Created a new partition 3 of type 'Linux' and of size 1 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device      Boot      Start        End Sectors  Size Id Type
/dev/sdb1                2048    1026047  1024000   500M 83 Linux
/dev/sdb2          1026048    3123199  2097152    1G 83 Linux
/dev/sdb3          3123200    5220351  2097152    1G 83 Linux

```

继续输入 n，建立扩展分区，分区类型已经默认为 e，所有选项回车，输入 p 查看分区表

```

Command (m for help): n
Partition type
  p   primary (3 primary, 0 extended, 1 free)
  e   extended (container for logical partitions)
Select (default e):

Using default response e.
Selected partition 4
First sector (5220352-10485759, default 5220352):
Last sector, +sectors or +size{K,M,G,T,P} (5220352-10485759, default 10485759):

Created a new partition 4 of type 'Extended' and of size 2.5 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device      Boot      Start        End Sectors  Size Id Type
/dev/sdb1                2048    1026047  1024000   500M 83 Linux
/dev/sdb2          1026048    3123199  2097152    1G 83 Linux
/dev/sdb3          3123200    5220351  2097152    1G 83 Linux
/dev/sdb4          5220352   10485759  5265408   2.5G  5 Extended

```

扩展分区不能直接使用，需要建立逻辑分区，输入 n，建立逻辑分区，其他默认回车，结束扇区指定 1G

输入 p 查看分区表情况

```

Command (m for help): n
All primary partitions are in use.
Adding logical partition 5
First sector (5222400-10485759, default 5222400): +^H^H^H
Value out of range.
First sector (5222400-10485759, default 5222400):
Last sector, +sectors or +size{K,M,G,T,P} (5222400-10485759, default 10485759): +1G

Created a new partition 5 of type 'Linux' and of size 1 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device      Boot      Start          End Sectors   Size Id Type
/dev/sdb1                2048    1026047  1024000   500M 83 Linux
/dev/sdb2          1026048    3123199  2097152    1G 83 Linux
/dev/sdb3          3123200    5220351  2097152    1G 83 Linux
/dev/sdb4          5220352   10485759  5265408   2.5G  5 Extended
/dev/sdb5          5222400    7319551  2097152    1G 83 Linux

```

继续建立逻辑分区，结束扇区回车，使用全部默认空间，输入 p 查看最后的分区信息

```

Command (m for help): n
All primary partitions are in use.
Adding logical partition 6
First sector (7321600-10485759, default 7321600):
Last sector, +sectors or +size{K,M,G,T,P} (7321600-10485759, default 10485759):

Created a new partition 6 of type 'Linux' and of size 1.5 GiB.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device      Boot      Start          End Sectors   Size Id Type
/dev/sdb1                2048    1026047  1024000   500M 83 Linux
/dev/sdb2          1026048    3123199  2097152    1G 83 Linux
/dev/sdb3          3123200    5220351  2097152    1G 83 Linux
/dev/sdb4          5220352   10485759  5265408   2.5G  5 Extended
/dev/sdb5          5222400    7319551  2097152    1G 83 Linux
/dev/sdb6          7321600   10485759  3164160   1.5G 83 Linux

```

最后，输入 w，将分区表同步到磁盘，完成分区，输入 q 退出 fdisk。

```

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

```

查看分区结果，使用 fdisk -l /dev/sdb 或者 lsblk

使用命令 lsblk 查看磁盘情况

```
mylinux@ubuntu:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sdb          8:16   0    5G  0 disk
├─sdb4       8:20   0    1K  0 part
├─sdb2       8:18   0    1G  0 part
├─sdb5       8:21   0    1G  0 part
├─sdb3       8:19   0    1G  0 part
├─sdb1       8:17   0   500M  0 part
└─sdb6       8:22   0   1.5G  0 part
sr0         11:0    1 1024M  0 rom
sda          8:0    0   20G  0 disk
├─sda2       8:2    0    1K  0 part
├─sda5       8:5    0   975M  0 part [SWAP]
└─sda1       8:1    0    19G  0 part /
```

注意：分区数量可以不到 4 个，但主分区和扩展分区总数不能超过 4 个，扩展分区没有或只有 1 个，扩展分区不能直接使用，需要建立逻辑分区才能存储数据

三、分区格式化

硬盘分区后，需要格式化，每种操作系统的文件属性和权限不同，需要格式化创建文件系统，使用 `mkfs.ext4 /dev/sdb1` 命令进行格式化。

使用方式：

`mkfs.` 文件类型（如 `ext4`） 分区名

`mkfs -t` 文件类型（如 `ext3`） 分区名

`mkfs` 分区名：默认 `ext2`

硬盘 `sdb` 和扩展分区不要格式化，

规划：

`sdb1 ext3`

`sdb3 ext4`

`sdb5 ext2`

将 `sdb1` 格式化为 `ext3` 文件系统，出现 `done` 后完成格式化


```
mylinux@ubuntu:~$ sudo mkfs -t ext3 /dev/sdb1
mke2fs 1.42.13 (17-May-2015)
Creating filesystem with 512000 1k blocks and 128016 inodes
Filesystem UUID: 71ee82e2-ab99-4462-ad4b-169133d2db78
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

将 sdb3 格式化为 ext4 文件系统，出现 done 后完成格式化

```
mylinux@ubuntu:~$ sudo mkfs.ext4 /dev/sdb3
mke2fs 1.42.13 (17-May-2015)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: c1afa22c-48b3-4909-82c7-06a5e8f4d8ff
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
```

将 sdb5 使用默认的文件系统格式化，默认为 ext2 文件系统，出现 done 后完成格式化

```
mylinux@ubuntu:~$ sudo mkfs /dev/sdb5
mke2fs 1.42.13 (17-May-2015)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: e6f49772-1081-4263-b0d7-e79b362e8a96
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Writing superblocks and filesystem accounting information: done
```

使用 blkid 命令查看格式化后的分区情况

```
mylinux@ubuntu:~$ sudo blkid
/dev/sda1: UUID="cb13fd40-447d-40c3-b6b3-2a9d91ac7ba3" TYPE="ext4" PARTUUID="b0eabb67-01"
/dev/sda5: UUID="3014e784-85d9-4d20-8c5f-923b230800a8" TYPE="swap" PARTUUID="b0eabb67-05"
/dev/sdb1: UUID="71ee82e2-ab99-4462-ad4b-169133d2db78" SEC_TYPE="ext2" TYPE="ext3" PARTUUID="aa11e55c-01"
/dev/sdb2: PARTUUID="aa11e55c-02"
/dev/sdb3: UUID="c1afa22c-48b3-4909-82c7-06a5e8f4d8ff" TYPE="ext4" PARTUUID="aa11e55c-03"
/dev/sdb5: UUID="e6f49772-1081-4263-b0d7-e79b362e8a96" TYPE="ext2" PARTUUID="aa11e55c-05"
/dev/sdb6: PARTUUID="aa11e55c-06"
```

四、挂载文件系统

文件系统挂载后才可以查看，默认挂载点/mnt 目录

挂载 mount，使用 `mkdir /mnt/disk1` 命令创建挂载点（注意：一个分区一个挂载点），使用 `mount /dev/sdb1 /mnt/disk1` 命令进行挂载。

创建/mnt 目录下 5 个挂载点目录

```
mylinux@ubuntu:~$ sudo mkdir -p /mnt/sdb{1..5}
mylinux@ubuntu:~$ cd /mnt
mylinux@ubuntu:/mnt$ ll
total 28
drwxr-xr-x  7 root root 4096 Nov 26 03:20 ./
drwxr-xr-x 23 root root 4096 Jul 28 18:15 ../
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb1/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb2/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb3/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb4/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb5/
```

查看/mnt 目录下的文件，使用 `mount` 命令将 `/dev/sdb1` 挂载到 `/mnt/sdb1` 目录下，完成后使用 `lsblk` 命令查看，显示挂载位置。

```
mylinux@ubuntu:/mnt$ ll
total 28
drwxr-xr-x  7 root root 4096 Nov 26 03:20 ./
drwxr-xr-x 23 root root 4096 Jul 28 18:15 ../
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb1/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb2/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb3/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb4/
drwxr-xr-x  2 root root 4096 Nov 26 03:20 sdb5/
mylinux@ubuntu:/mnt$ sudo mount /dev/sdb1 /mnt/sdb1
mylinux@ubuntu:/mnt$ lsblk
NAME      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
sdb        8:16   0    5G  0 disk
├─sdb4     8:20   0    1K  0 part
├─sdb2     8:18   0    1G  0 part
├─sdb5     8:21   0    1G  0 part
├─sdb3     8:19   0    1G  0 part
├─sdb1     8:17   0 500M  0 part /mnt/sdb1
└─sdb6     8:22   0  1.5G  0 part
```

挂载完成可以使用 `df -hT` 命令或 `mount` 命令查看挂载信息。

```
mylinux@ubuntu:/mnt$ df -Th
Filesystem      Type      Size  Used Avail Use% Mounted on
udev            devtmpfs  980M   0    980M   0% /dev
tmpfs           tmpfs     202M   6.2M 195M   4% /run
/dev/sda1       ext4      19G    3.9G 14G    22% /
tmpfs           tmpfs     1006M  152K 1006M   1% /dev/shm
tmpfs           tmpfs     5.0M   4.0K 5.0M   1% /run/lock
tmpfs           tmpfs     1006M   0    1006M   0% /sys/fs/cgroup
tmpfs           tmpfs     202M   16K  202M   1% /run/user/1000
/dev/sdb1       ext3      477M   2.3M 449M   1% /mnt/sdb1
```

进入挂载目录，建立文件保存

```
mylinux@ubuntu:/mnt/sdb3$ sudo vim file
mylinux@ubuntu:/mnt/sdb3$ ll
total 28
drwxr-xr-x 3 root root 4096 Nov 26 03:28 ./
drwxr-xr-x 7 root root 4096 Nov 26 03:20 ../
-rw-r--r-- 1 root root    6 Nov 26 03:28 file
```

可以使用 `umount` 命令卸载文件系统，卸载后将看不到该文件系统，但是文件内容还在分区中，挂载后又可以查看。

五、自动挂载

系统重启后，查看挂载点后，分区将无法使用，可以使用永久挂载。通过修改 `/etc/fstab` 文件，系统开机时检查该文件中的内容，自动挂载文件系统

```
mylinux@ubuntu:/mnt/sdb3$ sudo vim /etc/fstab
```

将 `sdb3` 设置为自动挂载，写入 `sdb3` 的 UUID (`blkid` 命令查看)，挂载点，文件类型

```
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=cb13fd40-447d-40c3-b6b3-2a9d91ac7ba3 / ext4 errors=remount-ro 0 1
# swap was on /dev/sda5 during installation
UUID=3014e784-85d9-4d20-8c5f-923b230800a8 none swap sw 0 0
UUID=clafa22c-48b3-4909-82c7-06a5e8f4d8ff /mnt/sdb3 ext4 defaults 0 0
```

保存退出后，一定执行命令 `mount -a` 检查语法格式。

```
mylinux@ubuntu:/mnt/sdb3$ sudo mount -a
```

六、交换分区

查看 `sdb` 分区情况，使用 `sdb2` 作为交换分区，现在 `sdb2` 的 Id 为 83


```
mylinux@ubuntu:/mnt/sdb3$ sudo fdisk -l /dev/sdb
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	1026047	1024000	500M	83	Linux
/dev/sdb2		1026048	3123199	2097152	1G	83	Linux
/dev/sdb3		3123200	5220351	2097152	1G	83	Linux
/dev/sdb4		5220352	10485759	5265408	2.5G	5	Extended
/dev/sdb5		5222400	7319551	2097152	1G	83	Linux
/dev/sdb6		7321600	10485759	3164160	1.5G	83	Linux

使用 fdisk 命令修改 id

```
mylinux@ubuntu:/mnt/sdb3$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.27.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): m

Help:

DOS (MBR)
 a  toggle a bootable flag
 b  edit nested BSD disklabel
 c  toggle the dos compatibility flag

Generic
 d  delete a partition
 F  list free unpartitioned space
 l  list known partition types
 n  add a new partition
 p  print the partition table
 t  change a partition type
 v  verify the partition table
 i  print information about a partition
```

输入命令 t，修改分区类型，输入要修改的分区号 2（sdb2），输入 1 查看分区类型，确认交换分区 Id 为 82

```

Command (m for help): t
Partition number (1-6, default 6): 2
Partition type (type L to list all types): l

0 Empty          24 NEC DOS          81 Minix / old Lin bf Solaris
1 FAT12          27 Hidden NTFS Win 82 Linux swap / So c1 DRDOS/sec (FAT-
2 XENIX root     39 Plan 9          83 Linux          c4 DRDOS/sec (FAT-
3 XENIX usr      3c PartitionMagic 84 OS/2 hidden or c6 DRDOS/sec (FAT-
4 FAT16 <32M     40 Venix 80286     85 Linux extended c7 Syrix
5 Extended       41 PPC PReP Boot  86 NTFS volume set da Non-FS data
6 FAT16          42 SFS            87 NTFS volume set db CP/M / CTOS / .
7 HPFS/NTFS/exFAT 4d QNX4.x          88 Linux plaintext de Dell Utility
8 AIX            4e QNX4.x 2nd part 8e Linux LVM       df BootIt
9 AIX bootable   4f QNX4.x 3rd part 93 Amoeba          e1 DOS access
a OS/2 Boot Manag 50 OnTrack DM     94 Amoeba BBT      e3 DOS R/O

```

输入 82，修改 sdb2 分区类型为交换分区

```

Partition type (type L to list all types): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.

```

输入 p，查看分区情况，sdb2 分区类型已经修改。

```

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c

Device      Boot      Start          End Sectors   Size Id Type
/dev/sdb1             2048     1026047 1024000    500M 83 Linux
/dev/sdb2          1026048     3123199 2097152      1G 82 Linux swap / Solaris
/dev/sdb3          3123200     5220351 2097152      1G 83 Linux
/dev/sdb4          5220352    10485759 5265408     2.5G  5 Extended
/dev/sdb5          5222400     7319551 2097152      1G 83 Linux
/dev/sdb6          7321600    10485759 3164160     1.5G 83 Linux

```

格式化交换分区 mkswap 命令

```

mylinux@ubuntu:/mnt/sdb3$ sudo mkswap /dev/sdb2
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=a8fd9dfe-deef-4c85-aabe-4695b2c488ee

```

使用 swapon -s 命令查看交换分区

```

mylinux@ubuntu:/mnt/sdb3$ swapon -s
Filename                                Type              Size      Used      Priority
/dev/sda5                              partition         998396      0         -2

```

使用 free -m 查看系统中的交换分区大小

```

mylinux@ubuntu:/mnt/sdb3$ free -m
              total        used         free       shared  buff/cache   available
Mem:           2011         185        1450          23         375        1574
Swap:           974           0          974

```

打开/dev/sdb2 交换分区后再查看

```
mylinux@ubuntu:/mnt/sdb3$ sudo swapon /dev/sdb2
mylinux@ubuntu:/mnt/sdb3$ sudo swapon -s
Filename                                Type              Size    Used    Priority
/dev/sda5                              partition         998396  0       -2
/dev/sdb2                              partition         1048572 0       -3
```

查看系统交换分区大小，已经发生变化

```
mylinux@ubuntu:/mnt/sdb3$ free -m
              total        used         free       shared  buff/cache   available
Mem:           2011         185         1450          23         375        1574
Swap:          1998           0         1998
```

交换分区也需要编辑/etc/fstab 永久挂载

将 sda5, sda6 建立逻辑卷

```
mylinux@ubuntu:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.27.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): m
```

```
Command (m for help): t
Partition number (1-6, default 6): 1
Partition type (type L to list all types): l

 0 Empty                24 NEC DOS               81 Minix / old Lin bf  Solaris
 1 FAT12                 27 Hidden NTFS Win 82  Linux swap / So c1  DRDOS/sec (FAT-
 2 XENIX root            39 Plan 9               83 Linux                c4  DRDOS/sec (FAT-
 3 XENIX usr              3c PartitionMagic      84 OS/2 hidden or c6  DRDOS/sec (FAT-
 4 FAT16 <32M            40 Venix 80286          85 Linux extended      c7  Syrinx
 5 Extended              41 PPC PReP Boot       86 NTFS volume set da  Non-FS data
 6 FAT16                 42 SFS                  87 NTFS volume set db  CP/M / CTOS / .
 7 HPFS/NTFS/exFAT       4d QNX4.x                88 Linux plaintext de  Dell Utility
 8 AIX                   4e QNX4.x 2nd part 8e  Linux LVM              df  BootIt
```

```
Partition type (type L to list all types): 8e

Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help): p
Disk /dev/sdb: 5 GiB, 5368709120 bytes, 10485760 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xaa11e55c
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	1026047	1024000	500M	8e	Linux LVM
/dev/sdb2		1026048	3123199	2097152	1G	82	Linux swap / Solaris
/dev/sdb3		3123200	5220351	2097152	1G	83	Linux
/dev/sdb4		5220352	10485759	5265408	2.5G	5	Extended
/dev/sdb5		5222400	7319551	2097152	1G	83	Linux
/dev/sdb6		7321600	10485759	3164160	1.5G	83	Linux


```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Re-reading the partition table failed.: Device or resource busy

The kernel still uses the old table. The new table will be used at the next reboot or after you run partprobe(
8) or kpartx(8).

mylinux@ubuntu:~$ partprobe
```

七、建立逻辑卷

1. 将 sdb5 和 sdb6 创建 pv

```
mylinux@ubuntu:~$ sudo pvcreate /dev/sdb5
WARNING: ext2 signature detected on /dev/sdb5 at offset 1080. Wipe it? [y/n]: y
Wiping ext2 signature on /dev/sdb5.
Physical volume "/dev/sdb5" successfully created
mylinux@ubuntu:~$ sudo pvcreate /dev/sdb6
Physical volume "/dev/sdb6" successfully created
```

2. 将 sdb5 和 sdb6 建立 vg1, 使用 vgcreate

```
mylinux@ubuntu:~$ sudo vgcreate vg1 /dev/sdb5 /dev/sdb6
Volume group "vg1" successfully created
```

3. 在 vg1 中建立 lv1, 使用 lvcreate

```
mylinux@ubuntu:~$ sudo lvcreate -L 200M -n lv1 vg1
Logical volume "lv1" created.
```

4. 格式化 lv1, 使用 mkfs

```
mylinux@ubuntu:~$ sudo mkfs.ext3 /dev/vg1/lv1
mke2fs 1.42.13 (17-May-2015)
Creating filesystem with 204800 1k blocks and 51200 inodes
Filesystem UUID: 8543321f-b12f-4bf5-aae5-19d826ba8f47
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done
```

5. 挂载 lv1 到 /dev/vg1/lv1, 使用 mount

```
mylinux@ubuntu:~$ sudo mkdir -p /mnt/vg1/lv1
mylinux@ubuntu:~$ sudo mount /dev/vg1/lv1 /mnt/vg1/lv1/
```